Reasons Why Narrowband IoT Offers Great Telecom Prospect

Due to an increase in connectivity demand, Narrowband IoT holds great potential for the upcoming market. IoT is therefore gradually shifting from wireless to narrowband

Dharmendra Kumar

The Internet of Things (IoT) collectively refers to an ecosystem of connected physical objects that are accessible through the Internet. Consisting of steps such as Collect, Communicate, Analyse, and Act, IoT interconnects three platforms: things, people, and cloud. In IoT, connectivity plays a major role. But since there are no defined rules for connectivity, we have to consider which kind of connectivity should be used: wired or wireless. Although wireless is convenient, it has a complex connectivity topology and protocols.

The four layers of IoT are:

1. Integrated applications. Any social media or GUI based app.

2. Information processing. Related to data gathering and interfacing with the integrated applications.

3. Network infrastructure. Related to how the network is going to be deployed for the end-user and what topology is exactly available. For the network topology of wireless communication, we have Wide Area Range, Low Power Wide Area Network, and Local Area Network.

4. Sensing and identification. Comprises sensors, which interface with networks for data communication.

The IoT reference diagram in Fig. 1 tells us how data will travel from the physical device to the last mile. As per the interface with the last mile and data requirement, relevant connectivity is required. Connectivity decides how much data is to be sent to the end-user and what kind of data packet is required for sending on the edge.

Some available IoT connectivity options are:

- Satellite communication
- Unlicensed band
- LTE - GSM band
- Narrowband IoT
- LoRa
- Local connectivity
- Wide area connectivity
- Low power wide area network (LPWAN)

Technologies such as LoRa, NBIoT, and Sigfox come under LPWAN. They have different use cases.

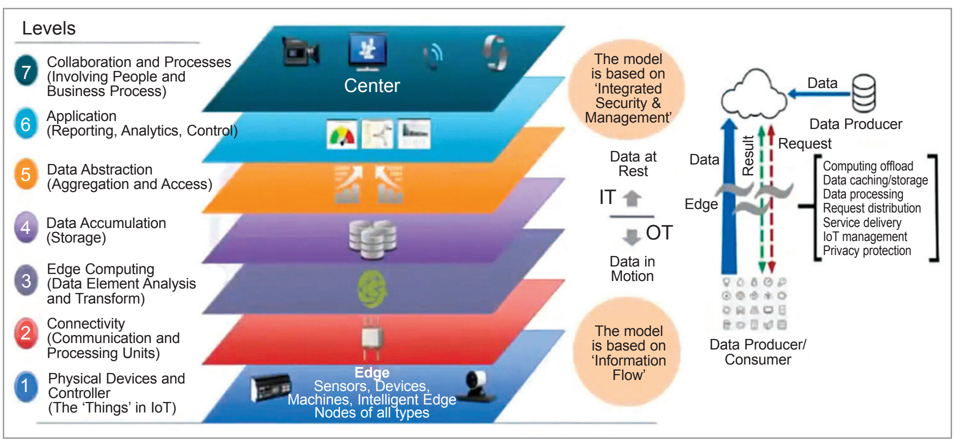

Fig. 1: IoT reference model

IoT connectivity demand

IoT connectivity is related to the kind of connectivity available on the device side as well as on the server side. Such connectivity types are:

Multi-sensor fusion. Het-